

Cryptography and Network Security

Dr. R.K. Chandrakar



Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Branch: B.Tech Honors Computer Science and Engineering (Artificial Intelligence)

Semester: V

Subject Code: C127573(022)

Subject: Cryptography and Network Security

Total Tutorial Periods: 10

Total Theory Periods: 40

No. of Assignments to be submitted: One per Unit

ESE Duration: Three Hours, Maximum Marks in ESE: 100

Minimum Marks in ESE: 35

UNIT-I: CLASSICAL ENCRYPTION TECHNIQUES

Symmetric cipher model, Steganography, Limitations of Perfect Secrecy, Shannon's Theorem.

UNIT-II: NUMBER THEORY

Prime numbers and factoring, modular arithmetic, computations in finite fields, Discrete logarithms.

UNIT-III: PRIVATE KEY ENCRYPTION SCHEMES AND BLOCK CIPHERS

Pseudorandom Functions and Permutations, Private-Key Encryption Schemes, DES – The Data Encryption Standard, Attacks on DES, Single-Round DES, Two-Round DES, Three-Round DES, Best Known Attacks on Increasing the Key size for DES, Modes of Operation

UNIT-IV: PUBLIC KEY ASYMMETRIC CRYPTOGRAPHY

Public-Key Problems and Mathematical Background, Diffie-Hellman Key Agreement, El-Gamal Encryption Scheme, RSA Encryption, Security of RSA, Hybrid Encryption, Attacks on RSA, Private and Public-Key, Timing Attacks, Elliptic Curve Cryptography.

UNIT-V: HASH FUNCTIONS

Definition and Properties, Constructions of Collision-Resistant Hash Functions, Popular Uses of Collision-Resistant Hash Functions, Random Oracle Model. Hash algorithms: SHA-512. Message Authentication: Message Authentication Codes Definitions, Constructions of Secure Message Authenticate Codes, Practical Constructions of Message Authentication Codes.

UNIT-VI: DIGITAL SIGNATURE AND APPLICATIONS AND SECURITY AT TRANSPORT LAYER

Definitions, and Public-Key Infrastructure, Combining Encryption and Signatures – SignCryption, SSL Architecture, Protocols, SSL Message formats, Transport Layer Security, Kerberos Password Management.

TEXT BOOKS:

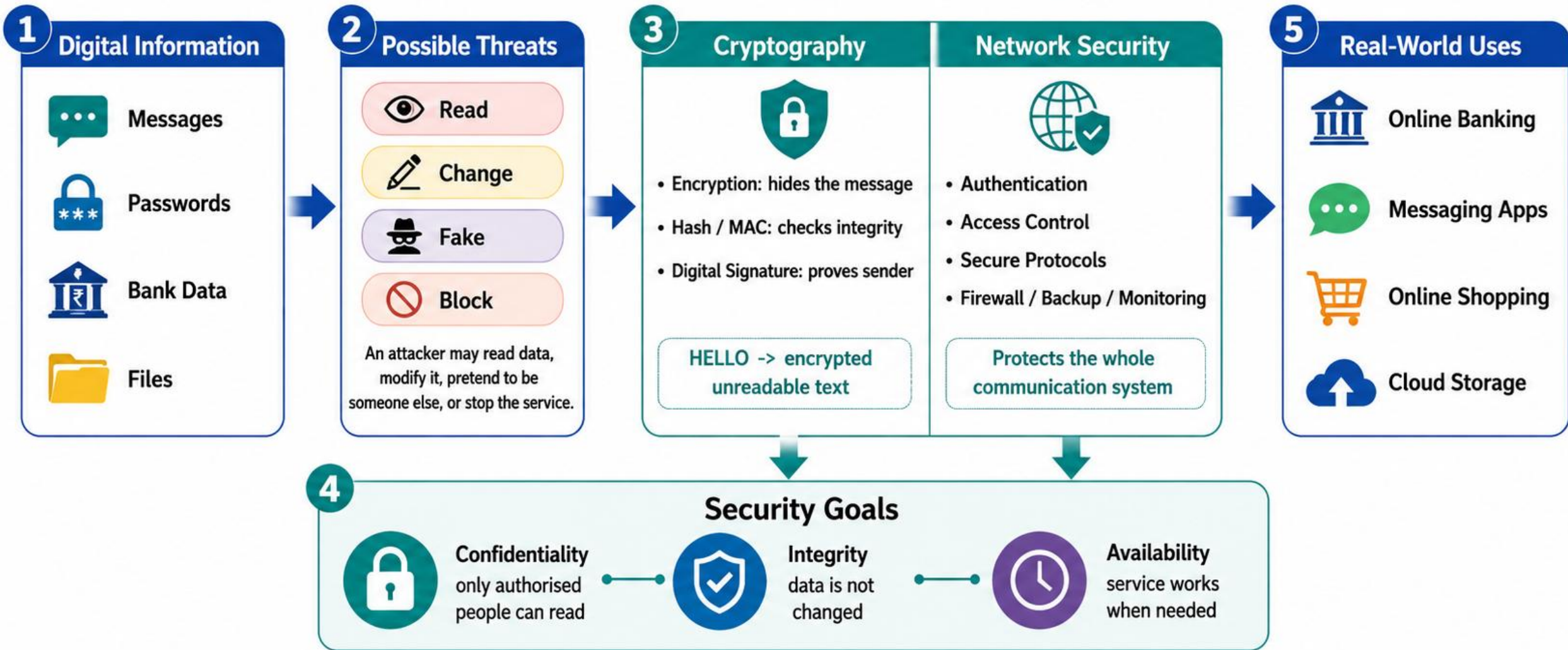
1. "Cryptography & Network Security" by William Stallings 4th Edition, 2006, Pearson Education Asia.
2. Kahate A, "Cryptography & Network Security", Tata McGraw Hill, 2004.

REFERENCE BOOKS:

1. "Applied Cryptology" by Schiner Bruce, John Wiley & Sons, 2001.
2. "Introduction to Cryptography with Coding Theory" by Wade Trappe & Lawrence C Washington, New Jersey, Pearson Education, 2006.
3. Charlie Kaufman, Radia Perlman and Mike Speciner, "Network Security: Private Communication in a Public World", Prentice Hall of India Private Limited.
4. Behrouz A. Forouzan, "Cryptography and Network Security", McGrawHill

Cryptography and Network Security: Basic Concept

What are we protecting, from whom, and how?



Cryptography protects information mathematically. **Network security** protects the complete system.

Before beginning the first unit, we must understand why this subject exists. Modern society depends on digital information.

Every day, we use digital systems for banking, messaging, online shopping, cloud storage, examinations, healthcare and government services. These systems continuously store and transmit sensitive information.

Situation	What must be protected?	What can go wrong?
Online banking	Login details, transaction amount and account balance	An attacker may steal credentials or change the payment destination.
Messaging app	Message content and identity of the sender	Someone may read the chat, impersonate a friend or alter a message.
University portal	Marks, attendance and student records	Marks may be modified, or the portal may become unavailable during registration.
Cloud storage	Files, access permissions and backups	Private files may leak or be deleted by ransomware.

Whenever information moves through a computer network, several questions arise:

- Can an unauthorized person read it?
- Can someone secretly modify it?
- How can we verify the identity of the sender?
- How can we prove that a message is genuine?
- How can two people communicate securely over an insecure network?
- What happens if an attacker records and reuses an old message?

The subject **Cryptography and Network Security** studies the techniques used to answer these questions.

Consider a student paying university fees through an online portal.
The system must ensure:

Security requirement	Meaning in this example
Confidentiality	Account and payment details must remain private.
Integrity	The payment amount and recipient must not be altered.
Authentication	The student and university portal must verify each other.
Freshness	An old payment request must not be accepted again.
Availability	The portal must remain accessible during payment.
Non-repudiation	A valid record of the completed payment should exist.

What is Cryptography?

- Cryptography is the science of protecting information by transforming it using mathematical algorithms and keys. The word comes from Greek roots meaning “hidden writing.”
- A readable message is called **plaintext**. After encryption, it becomes an unreadable form called **ciphertext**.

$$\text{Plaintext} \xrightarrow[\text{Key}]{\text{Encryption}} \text{Ciphertext}$$

- The authorized receiver applies decryption to recover the original message.

$$\text{Ciphertext} \xrightarrow[\text{Key}]{\text{Decryption}} \text{Plaintext}$$

- Cryptography is not limited to hiding information. It is also used to verify identities, detect changes in data and provide proof of digital actions.

Cryptography Core terminology

Term	Meaning	Example
Plaintext (P)	Original readable data.	MEET AT 5 PM
Encryption (E)	Process that converts plaintext into ciphertext.	Apply Caesar shift +3.
Ciphertext (C)	Scrambled output that should not reveal the message.	PHHW DW 5 SP
Decryption (D)	Process that recovers plaintext from ciphertext.	Apply shift -3.
Algorithm / Cipher	The rule or procedure used for encryption and decryption.	Caesar cipher, AES(Advanced Encryption Standard).
Key (K)	A value that controls the transformation.	Shift value 3, or a 128-bit AES key.
Cryptanalysis	Study of recovering information without knowing the secret key.	Frequency analysis on a substitution cipher.
Cryptology	The combined field of cryptography and cryptanalysis.	Designing and analyzing secure ciphers.

Historical Background

Cryptography did not begin with computers. It developed from military, diplomatic and commercial needs. Classical methods mostly work on letters or symbols; modern methods work on bits and use advanced mathematics.

System	Easy meaning	Simple example
Ancient transposition	Mix the positions of letters	TEAM → TEMA
Caesar cipher	Move every letter by the same number	CAT → FDW using +3
Polyalphabetic cipher	Move each letter by a different number	CAT → DCW using +1, +2, +3
Rotor machine	The machine changes the rule after every letter	AAA → QXM because the rotor keeps turning
Computer-era cipher	A computer locks digital data using a secret key	HELLO → encrypted unreadable data using AES

Main Goals of the Security

The first three goals are known as the **CIA triad**.

➤ Confidentiality

Confidentiality ensures that information is visible only to authorized users.

Example: Only the account owner should be able to see a bank statement.

Common mechanisms include encryption, passwords and access control.

➤ Integrity

Integrity ensures that information has not been changed without permission.

Example: A payment amount of ₹1,000 must not secretly become ₹10,000.

Common mechanisms include hashes, message authentication codes and digital signatures.

➤ Availability

Availability ensures that authorized users can access systems and information when required.

Example: A university examination portal should remain available during registration and result publication.

Common mechanisms include backups, redundancy and denial-of-service protection.

What Is Network Security?

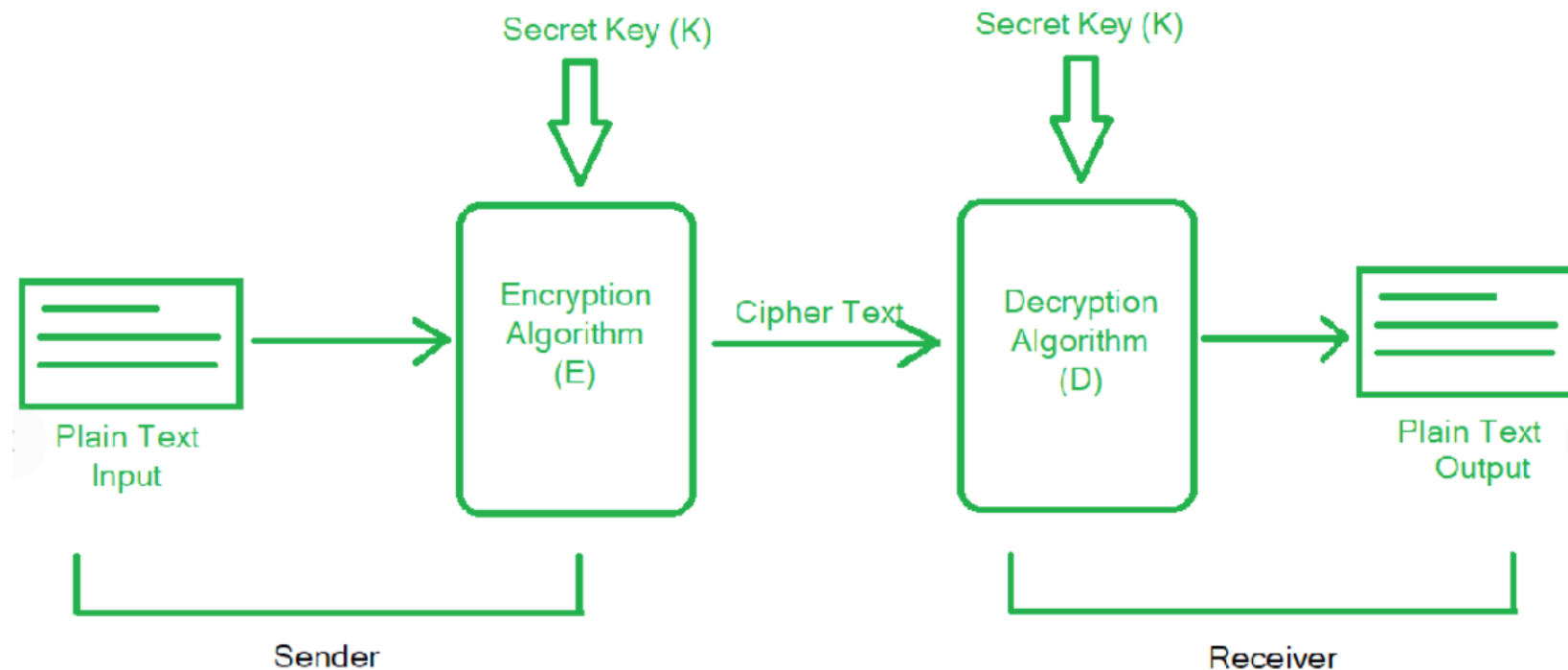
- Network security is the protection of information, devices, services and communication channels from unauthorized access, modification, misuse and disruption.
- Cryptography provides important mathematical tools, but complete network security also requires:
 - **Authentication:** The process of verifying that a person, device or message is really who or what it claims to be.
Example: Entering a password or OTP to prove you are the account owner.
 - **Access Control:** The process of allowing **only authorized users to access specific resources or perform specific actions.**
Example: A student can view their marks, but only the teacher can edit them.
 - **Firewall:** A security system that **monitors and controls incoming and outgoing network traffic** based on predefined rules.
Example: A firewall can block an unknown computer from accessing your university's internal network.
 - **Secure Protocols:** **Rules and procedures that protect data while it is being transmitted over a network.**
Example: **HTTPS** protects information exchanged between your browser and a website.
 - **Software Security:** The practice of protecting software from **bugs, vulnerabilities, and unauthorized access or misuse.**
Example: A banking app checks user input and uses secure coding to prevent attackers from stealing account information.

Therefore:

Cryptography protects information mathematically, while network security protects the complete communication system.

The symmetric cipher model

- The symmetric cipher model is a basic cryptography model where the **same secret key** is used to encrypt plaintext and decrypt ciphertext.
- It is fast and efficient, but the key must be shared securely between sender and receiver.
- In this model, both the sender and the receiver possess the same secret key (K).

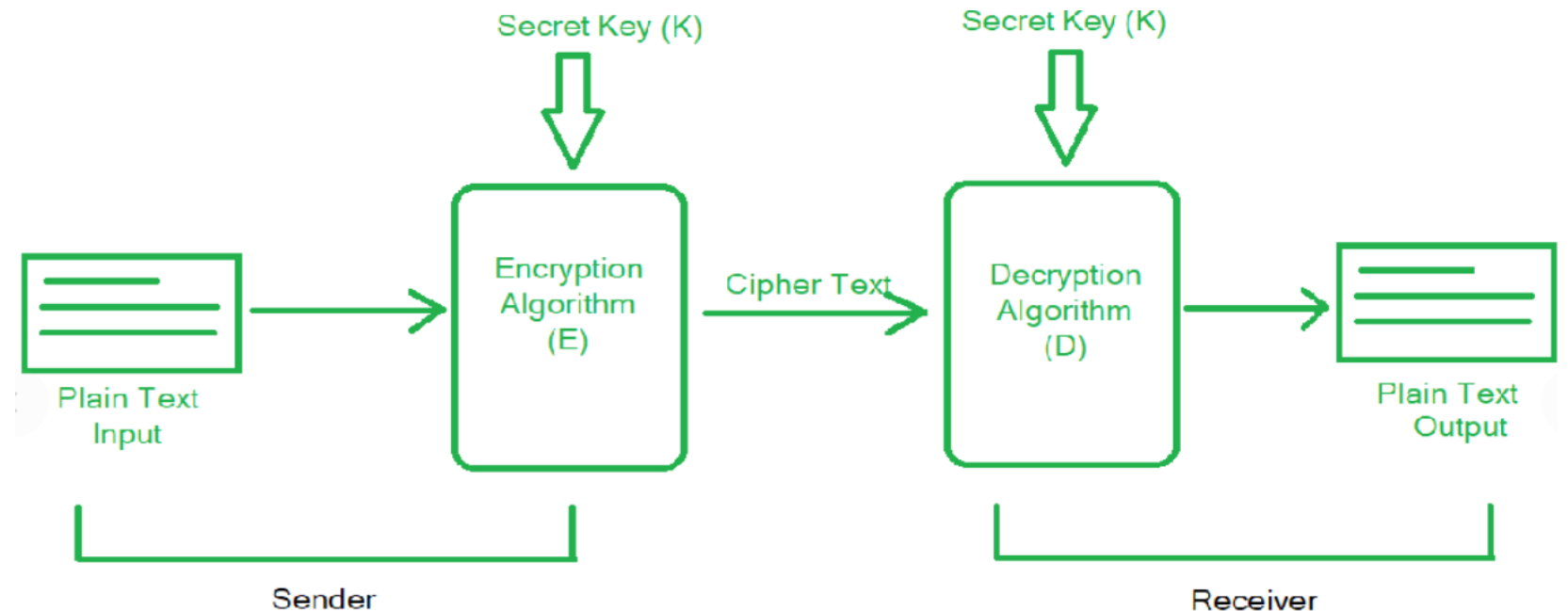


Components of the Symmetric Cipher Model

- **Plaintext (P):** the original readable message.
- **Encryption algorithm (E):** converts plaintext into ciphertext using the secret key.
- **Secret key (K):** shared between sender and receiver.
- **Ciphertext (C):** the encrypted message transmitted over the communication channel.
- **Decryption algorithm (D):** converts ciphertext back into plaintext using the same key.

Working process

1. The sender writes the plaintext message.
2. The encryption algorithm uses the secret key K to encrypt the plaintext.
3. The ciphertext is transmitted through an insecure channel.
4. The receiver applies the decryption algorithm using the same secret key K .
5. The original plaintext is recovered.



Mathematical representation

- $y = E(x, k)$: encrypt plaintext x with key k to get ciphertext y .
- $x = D(y, k)$: decrypt ciphertext y with the same key k to recover plaintext x .

Example

- Suppose:
- Plaintext = HELLO
- Secret key = 3
- Using a Caesar cipher:
 - $H \rightarrow K$
 - $E \rightarrow H$
 - $L \rightarrow O$
 - $L \rightarrow O$
 - $O \rightarrow R$
- Ciphertext = KHOOR
- The receiver uses the same key (3) to decrypt KHOOR back to HELLO.

Characteristics

- Uses the same secret key for encryption and decryption.
- The sender and receiver must both know the shared key before communication begins.
- It is best for bulk data encryption because it is fast and efficient.
- Security depends heavily on keeping the key secret and managing it well.

Advantages

- Fast and efficient for encrypting large amounts of data.
- Simple to understand and implement.
- Well suited for real-time communication, databases, backups and similar large-scale use.
- Uses less computational power and memory than asymmetric encryption.

Disadvantages

- Key distribution is difficult because the shared key must be exchanged securely.
- If the key is stolen or intercepted, the data is compromised.
- Each pair of users needs its own key, so key management becomes complex as users increase.

Common symmetric algorithms

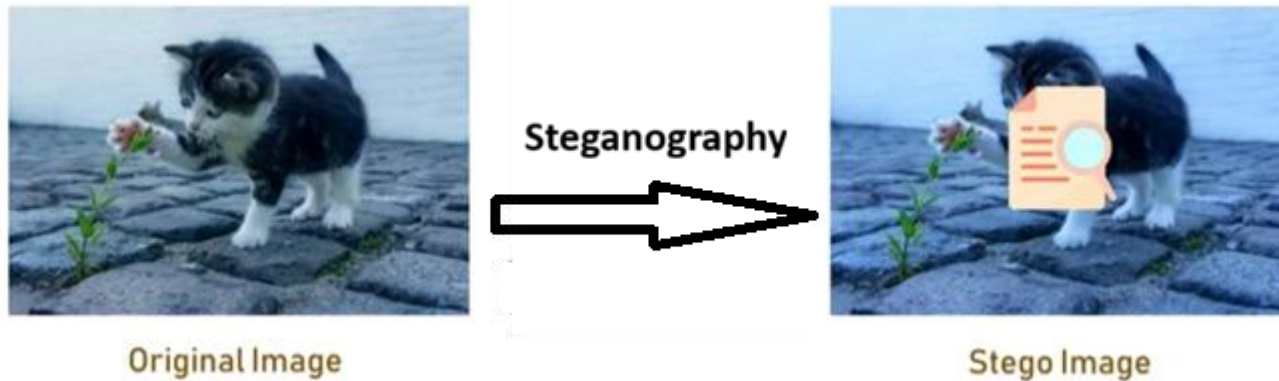
- AES (Advanced Encryption Standard) — most widely used.
- DES (Data Encryption Standard) — obsolete.
- 3DES (Triple DES).
- Blowfish.
- Twofish.

Symmetric vs asymmetric encryption

Feature	Symmetric encryption	Asymmetric encryption
Keys used	One shared secret key	Two keys: public key and private key
Speed	Faster and more efficient	Slower and more computationally expensive
Key sharing	Harder, because the secret key must be shared securely	Easier, because the public key can be shared openly
Main use	Bulk data encryption, fast communication	Secure key exchange, digital signatures, authentication
Security benefit	Efficient, but depends heavily on safe key distribution	Better for identity and trust features like signatures

Steganography

- Steganography is the practice of hiding the existence of a message inside another file or medium, such as an image, audio, video, or text, so that it does not look secret at first glance.
- It is different from encryption, which hides the content of a message; steganography hides the fact that a message exists.

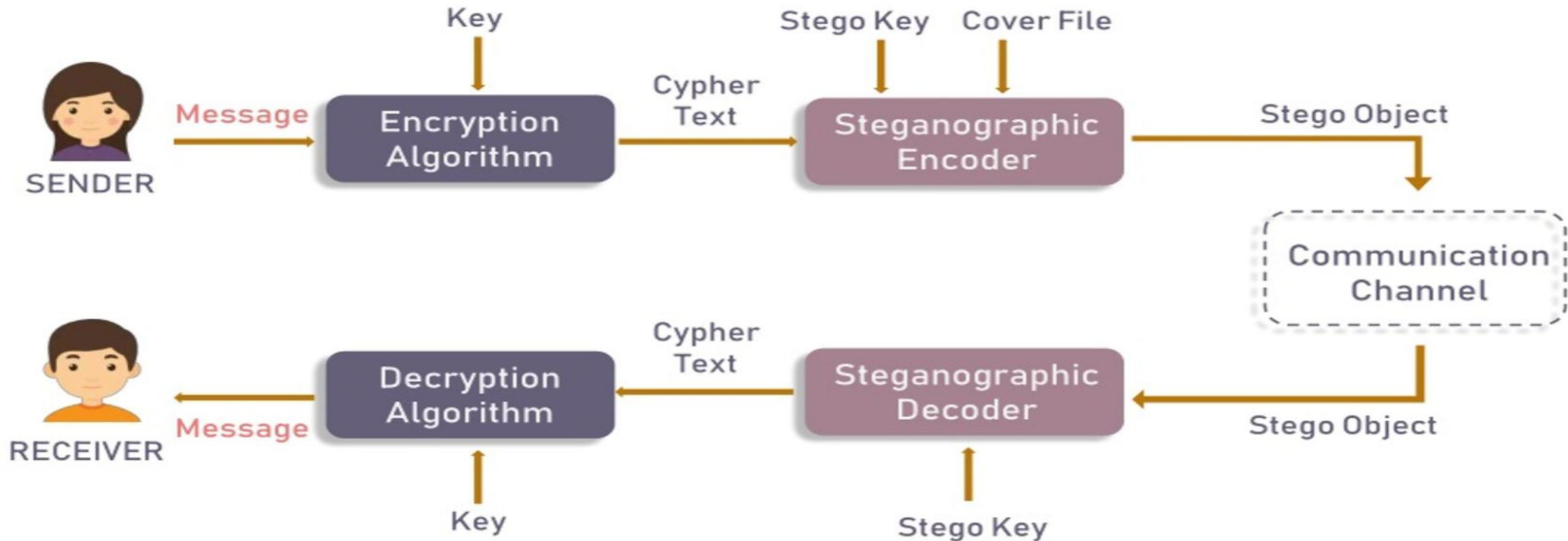


Components of steganography

- **Cover object:** the original image, audio, video or text used to hide the message.
- **Secret message:** the confidential information to be hidden.
- **Stego key (optional):** a secret key used during embedding and extraction.
- **Stego object:** the output file containing the hidden message.

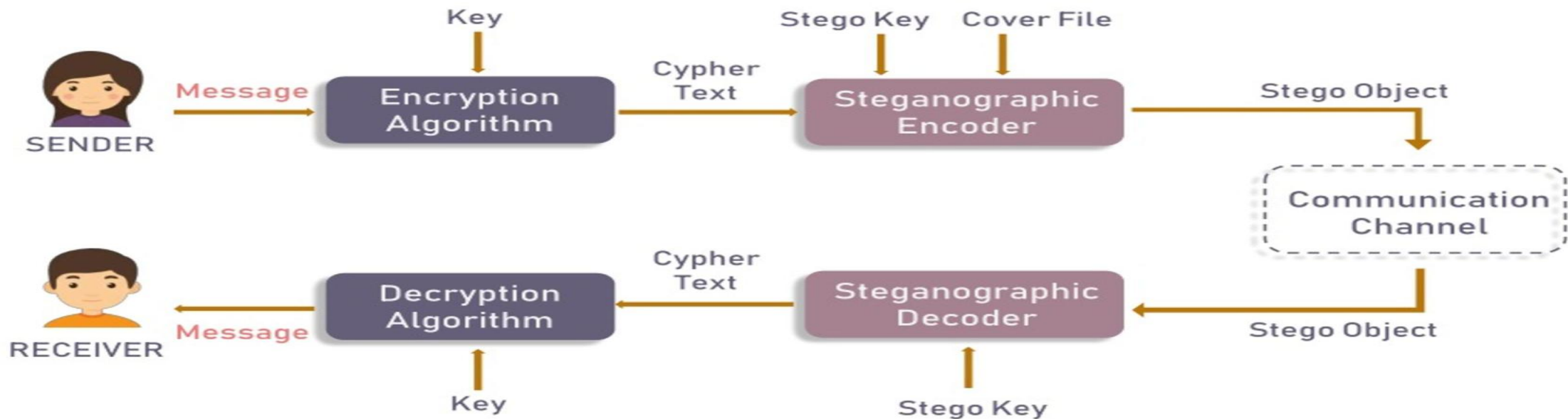
Working principle

1. Sender prepares a cover image and a secret message.
2. The embedding algorithm hides the secret message inside the cover image using a stego key.



Working principle

3. The result is a stego image, which looks like a normal image but contains the hidden message.
4. The receiver uses the extraction algorithm and the same stego key to recover the hidden message.
5. Finally, the recovered secret message is obtained.



Mathematical Representation

Embedding:

$$\text{Stego Object} = \text{Embed}(\text{Cover Object}, \text{Secret Message}, \text{Key})$$

Extraction:

$$\text{Secret Message} = \text{Extract}(\text{Stego Object}, \text{Key})$$

Types of Steganography

1. Text steganography

The message is hidden in a text document using spaces, formatting, punctuation or character positions.

2. Image steganography

The message is hidden in image pixels. This is the most common type.

3. Audio steganography

The message is embedded in audio signals by modifying sound samples.

4. Video steganography

The message is hidden in video frames or audio tracks.

Advantages

- Hides the existence of the message.
- Provides an additional layer of security.
- High hiding capacity in multimedia files.
- Difficult for attackers to detect visually.
- Useful for copyright protection and watermarking.

Disadvantages

- Hidden data may be destroyed by compression or editing.
- Limited embedding capacity.
- If detected, the hidden message may be extracted.

Steganography vs cryptography

Steganography	Cryptography
Hides the existence of the message	Hides the content of the message
Uses a cover object	Uses an encryption algorithm
Output looks ordinary	Output looks encrypted
Difficult to detect	Easy to detect but hard to read

Limitation

- Steganography does not make data unreadable by itself, so it is often used together with encryption for better security.

Applications

- Secure military communication.
- Digital watermarking.
- Copyright protection.
- Medical data hiding.
- Covert communication.
- Authentication systems.

Perfect Secrecy

- Perfect secrecy is a property of an encryption system in which the ciphertext provides no information about the plaintext. Even if an attacker has unlimited computational power and knows the encryption algorithm, they cannot determine the original message from the ciphertext without the key.
- This concept was introduced by Claude Shannon in 1949.
- The key idea is that the ciphertext and plaintext are statistically independent.
- In other words, after seeing the encrypted message, an attacker's belief about the original message does not change.

Perfect Secrecy Model

Sender

Plaintext + secret key

Encryption algorithm

Produces ciphertext using the secret key.

Ciphertext

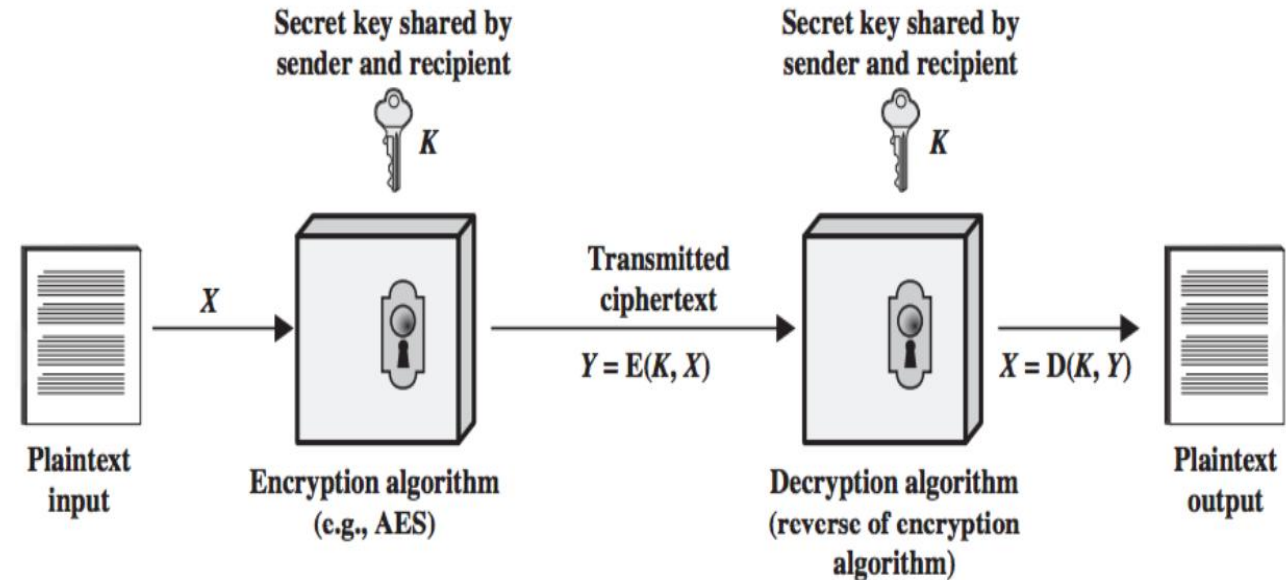
Intercepted by the attacker, but reveals no information about the plaintext.

Decryption algorithm

Uses the same secret key to recover the plaintext.

Receiver

Recovered plaintext.



Perfect Secrecy

An encryption system has perfect secrecy if

$$P(M = m \mid C = c) = P(M = m)$$

where:

M = plaintext (message),

C = ciphertext,

m = a particular plaintext,

c = a particular ciphertext.

This means the probability of any message after seeing the ciphertext is exactly the same as before seeing the ciphertext.

Suppose a ciphertext is intercepted.

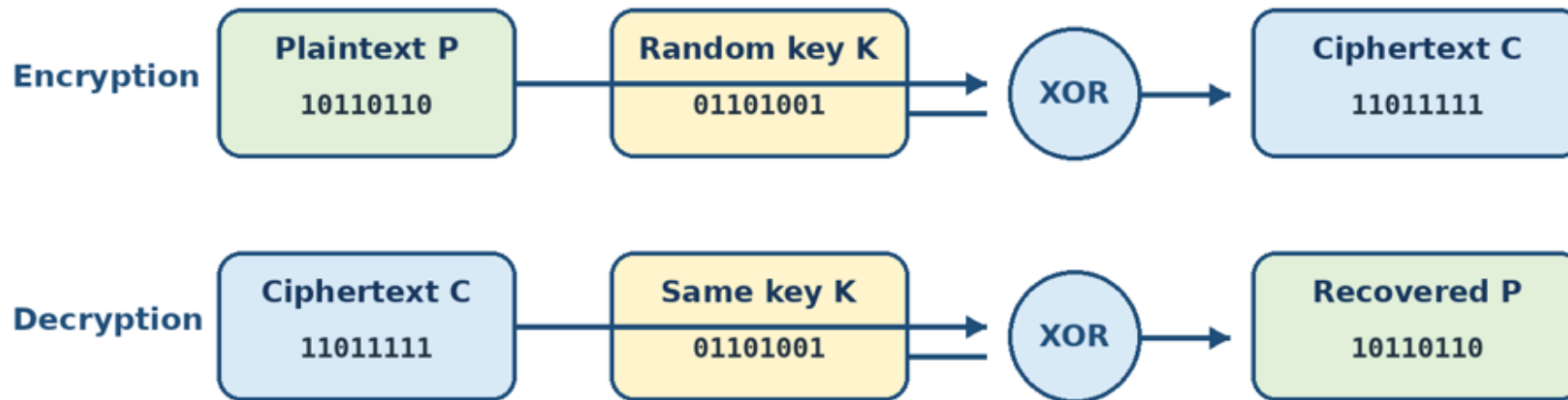
- Before interception, an attacker believes message A and message B are equally likely.
- After interception, if the attacker still considers both messages equally likely, then the encryption achieves perfect secrecy.

The ciphertext does not help the attacker at all.

The one-time pad

The one-time pad (OTP) is the standard construction that can achieve perfect secrecy. For binary data it combines the plaintext and key using XOR. XOR is reversible: applying the same key again returns the original plaintext.

One-Time Pad (OTP): XOR model



- $C = P \text{ XOR } K$, and $P = C \text{ XOR } K$.
- For perfect secrecy, K must be truly random, as long as P , secret, and never reused.

Conditions required for perfect secrecy with an OTP

All conditions must hold - missing one breaks the guarantee.

Shannon proved that perfect secrecy is achieved only if the following conditions are satisfied.

- **Random key**

The key must be generated uniformly at random.

- **Key length**

The key length must be at least equal to the message length.

- **One-time use**

Each key must be used only once.

- **Secret key**

The key must remain completely secret.

Limitations of Perfect Secrecy

- The key must be at least as long as the message.
- The key must be truly random.
- The key must never be reused.
- Secure key distribution is difficult.
- Key storage and management are complicated.
- It is inefficient for large-scale communication.

Perfect Secrecy VS Computational Security

Aspect	Perfect secrecy	Computational security
Security goal	Ciphertext reveals no information about plaintext.	Ciphertext is secure against efficient attackers .
Attacker model	Even an attacker with unlimited power cannot learn anything.	Security holds only for attackers with limited computational resources.
Guarantee type	Information-theoretic guarantee.	Practical, complexity-based guarantee.
Key requirement	Usually needs key as long as the message and fresh keys.	Can use shorter, reusable keys.
Real-world use	Rare in practice; one-time pad is the classic example.	Used in modern cryptography.

Shannon's Theorem on Perfect Secrecy

- Shannon's theorem on perfect secrecy states that an encryption system achieves perfect secrecy only if the uncertainty (entropy) of the key is at least equal to the uncertainty of the message.
- Claude Shannon introduced this concept in his 1949 paper Communication Theory of Secrecy Systems.
- The theorem establishes the necessary condition for achieving perfect secrecy.

Shannon's Theorem on Perfect Secrecy

- An encryption system provides perfect secrecy if the ciphertext reveals no information about the plaintext.

- Mathematically,

$$P(M = m \mid C = c) = P(M = m)$$

- for every plaintext m and ciphertext c .
- This theorem explains why perfect secrecy is hard to achieve in practice: the key must be truly random, as long as the message, and used only once in schemes like the one-time pad.

Shannon's Entropy Condition

The theorem states that

$$H(K) \geq H(M)$$

Where:

$H(K)$ = entropy or uncertainty of the key

$H(M)$ = entropy or uncertainty of the message

$\geq$ = greater than or equal to

Easy meaning: The key must be at least as unpredictable as the message.

Example: For an unrestricted **100-bit message**, perfect secrecy requires at least **100 bits of secret-key entropy**.

A shorter key may provide strong computational security, but it cannot provide perfect secrecy for every possible message.

Shannon's theorem (necessary and sufficient form)

For perfect secrecy, the following conditions must hold:

1. The key is chosen uniformly at random.
2. The key is at least as long as the message.
3. Each key is used only once.
4. The key remains completely secret.

These conditions are satisfied by the one-time pad (OTP).

Importance of the theorem

- Shannon's theorem shows that perfect secrecy has a fundamental cost: large random keys.
- It proves that no encryption algorithm can achieve perfect secrecy with a short reusable key.
- This is why practical systems such as AES use computational security instead of perfect secrecy.

Advantages

- Absolute confidentiality.
- Independent of computational power.
- Strongest known security guarantee.

Limitations

- Large key storage requirement.
- Difficult key distribution.
- Keys cannot be reused.
- Impractical for large-scale communication.